

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397622
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Johnson; David K. et al.

Vehicular rear slider window assembly with anti-vibration elements on movable window panel

Abstract

A vehicular rear slider window assembly includes a fixed window panel, a movable window panel and a frame portion. An upper edge region of the movable window panel is movably received in a channel of an upper rail of the frame portion. A first biasing element is compressed between a first inner surface of the upper rail and a first side of the movable window panel and urges the movable window panel away from the first inner surface. A second biasing element is compressed between a second inner surface of the upper rail and a second side of the movable window panel and urges the movable window panel away from the second inner surface. As the movable window panel is moved between the closed and opened positions, the first and second biasing elements guide movement of the movable window panel in a direction parallel to a centerline of the channel.

Inventors:	Johnson; David K. (Holland, MI), Bowman; Kyle S. (Holland, MI)
Applicant:	Magna Exteriors, Inc. (Troy, MI)
Family ID:	1000008778402
Assignee:	Magna Exteriors, Inc. (Troy, MI)
Appl. No.:	18/670792
Filed:	May 22, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240391303 A1	Nov. 28, 2024

Related U.S. Application Data

us-provisional-application US 63504454 20230526

Publication Classification

Int. Cl.: B60J1/18 (20060101)

U.S. Cl.:

CPC B60J1/1853 (20130101);

Field of Classification Search

CPC: B60J (1/1853)

USPC: 49/413; 49/414; 49/415; 49/416; 16/DIG.6

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2132104	12/1937	Froeliger	49/440	B60J 10/74
2209041	12/1939	Sinchuk	292/77	E06B 3/44
2781876	12/1956	Van Fleet	292/76	E06B 3/44
4004629	12/1976	Kelly	160/90	E06B 3/4681
5294168	12/1993	Kronbetter	296/146.16	E06B 3/4681
5473840	12/1994	Gillen	49/408	B60J 10/74
5505023	12/1995	Gillen	49/408	B60J 1/1853
5771637	12/1997	Oikawa	49/416	E05F 11/382
5799444	12/1997	Freimark et al.	N/A	N/A
5836110	12/1997	Buening	74/89.22	E05F 11/505
5943823	12/1998	Yoshida	49/416	B60J 5/0402
6305125	12/2000	Nozaki	49/419	B60J 1/17
6691464	12/2003	Nestell et al.	N/A	N/A
7003916	12/2005	Nestell et al.	N/A	N/A
7062880	12/2005	Renke	49/416	B60J 1/17
7073293	12/2005	Galer	N/A	N/A
8402695	12/2012	Smith et al.	N/A	N/A
8881458	12/2013	Snider	49/213	E05C 1/08
8915018	12/2013	Snider	N/A	N/A
8938914	12/2014	Hulst et al.	N/A	N/A
9266409	12/2015	Tooker et al.	N/A	N/A
9579955	12/2016	Snider	N/A	N/A
9731580	12/2016	Snider et al.	N/A	N/A
10023026	12/2017	Snider et al.	N/A	N/A
10501977	12/2018	Snider et al.	N/A	N/A
10841983	12/2019	Snider	N/A	B60J 1/002
11247546	12/2021	Tooker et al.	N/A	N/A
11331984	12/2021	Gustafson	N/A	N/A
11912110	12/2023	Snider et al.	N/A	N/A
11952820	12/2023	Snider et al.	N/A	N/A
12043092	12/2023	Tooker et al.	N/A	N/A

12065023	12/2023	Snider et al.	N/A	N/A
2003/0213179	12/2002	Galer	N/A	N/A
2004/0020131	12/2003	Galer et al.	N/A	N/A
2006/0107600	12/2005	Nestell et al.	N/A	N/A
2008/0127563	12/2007	Tooker	N/A	N/A
2008/0209814	12/2007	Tarjeft	49/415	B60J 5/0402
2012/0091114	12/2011	Ackerman	49/70	B60J 1/1853
2012/0117880	12/2011	Lahnala	49/70	B60J 1/1853
2014/0047772	12/2013	Hulst	N/A	N/A
2018/0010374	12/2017	Gipson	N/A	B60J 1/1853
2019/0383084	12/2018	Snider et al.	N/A	N/A
2021/0355737	12/2020	Snider et al.	N/A	N/A
2024/0391303	12/2023	Johnson	N/A	B60J 1/1853

Primary Examiner: Sosnowski; David E

Attorney, Agent or Firm: HONIGMAN LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application claims the filing benefits of U.S. provisional application Ser. No. 63/504,454, filed May 26, 2023, which is hereby incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

(1) The present invention relates to a slider window assembly for a vehicle and, more particularly, a rear slider window assembly for a vehicle and, more particularly, a rear slider window assembly for a pickup truck or the like.

BACKGROUND OF THE INVENTION

(2) It is known to provide a slider window assembly for an opening of a vehicle, such as a rear slider window assembly for a rear opening of a pickup truck. Conventional slider window assemblies for rear openings of trucks typically include three or more panels, such as two fixed window panels and a slidable window panel. The slidable window panel is supported by rails and may be moved along the rails to open and close the window.

SUMMARY OF THE INVENTION

(3) A vehicular rear window slider assembly includes a fixed window panel, a movable window panel and a frame portion fixed relative to the fixed window panel. The frame portion includes an upper rail. The fixed window panel includes an opening, and the movable window panel is movable along the upper rail between (i) a closed position, where the movable window panel is disposed at the opening, and (ii) an opened position, where the movable window panel is disposed at least partially along the fixed window panel. An upper edge region of the movable window panel is movably received in a channel of the upper rail. A first biasing element is disposed between a first side of the movable window panel along the upper edge region and a first inner surface of the upper rail along the channel. A second biasing element is disposed between a second side of the movable window panel along the upper edge region and a second inner surface of the upper rail along the channel. The second side of the movable window panel is opposite the first side. The first inner surface and the second inner surface are along opposing sides of the channel. The first biasing element is at least partially compressed between the first inner surface of the upper rail and the first side of the movable window panel and urges the movable window panel away from the first inner

surface of the upper rail. The second biasing element is at least partially compressed between the second inner surface of the upper rail and the second side of the movable window panel and urges the movable window panel away from the second inner surface of the upper rail.

(4) These and other objects, advantages, purposes and features of the present invention will become apparent upon review of the following specification in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a rear perspective view of a pickup truck having a rear slider window assembly;
- (2) FIG. 2 is a plan view of a rear slider window assembly, where a guide element having biasing elements is attached along an upper edge region of a movable window panel and the guide element and upper edge region of the movable window panel are movably received along an upper rail of a frame of the rear slider window assembly;
- (3) FIG. 2A is a sectional view taken along line A-A in FIG. 2;
- (4) FIG. 2B is a sectional view taken along line B-B in FIG. 2;
- (5) FIG. 2C is a sectional view taken along line C-C in FIG. 2;
- (6) FIG. 3 is a side view of the rear slider window assembly of FIG. 2;
- (7) FIG. 3A is a sectional view taken along line A-A in FIG. 3;
- (8) FIG. 4 is an exploded view of the movable window panel, guide element and upper rail of the rear slider window assembly of FIG. 2;
- (9) FIG. 5 is a perspective view of the movable window panel with the attached guide element having biasing elements of the rear slider window assembly of FIG. 2;
- (10) FIGS. 6-9 are views of the guide element having biasing elements of the rear slider window assembly of FIG. 2;
- (11) FIG. 10 is a plan view of another rear slider window assembly, where biasing elements are attached along an upper edge region of a movable window panel and the biasing elements and upper edge region of the movable window panel are movably received along an upper rail of a frame of the rear slider window assembly;
- (12) FIG. 10A is a sectional view taken along line A-A in FIG. 10;
- (13) FIG. 10B is a sectional view taken along line B-B in FIG. 10;
- (14) FIG. 10C is a sectional view taken along line C-C in FIG. 10;
- (15) FIG. 11 is a side view of the rear slider window assembly of FIG. 10;
- (16) FIG. 11A is a sectional view taken along line A-A in FIG. 11;
- (17) FIG. 12 is an exploded view of the movable window panel, biasing elements and upper rail of the rear slider window assembly of FIG. 10;
- (18) FIG. 13 is a perspective view of the movable window panel with the attached biasing elements of the rear slider window assembly of FIG. 10; and
- (19) FIGS. 14-17 are views of one of the biasing elements of the rear slider window assembly of FIG. 10.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(20) Referring now to the drawings and the illustrative embodiments depicted therein, a rear slider window assembly 10 of a vehicle 12 (such as a pickup truck or the like) includes a window frame 14, a fixed window panel or panels having an aperture that separates side window panels or portions 16, 18 and a movable window panel 20 that is movable relative to the window frame 14 and the fixed window panel between a closed position, where the movable window panel 20 is disposed at the aperture of the fixed window panel, and an opened position, where the movable window panel 20 is moved at least partially along the fixed window panel (FIG. 1).

(21) The movable window panel 20 may be movable along the frame 14 (such as along an upper

rail or channel and/or a lower rail or channel of the frame) relative to the fixed window panel via a drive system that moves the movable window panel in either direction to open and close the rear window, such as responsive to a user input or button or switch in the vehicle cabin. Optionally, the window panel **20** is manually movable along the frame **14** relative to the fixed window panel between the opened position and the closed position.

(22) As discussed further below, a biasing or absorbing element or damping element or attenuating element is disposed along at least one of the upper edge region and the lower edge region of the movable window panel **20** and received along the corresponding upper rail or lower rail of the frame. The biasing element is at least partially compressed or flexed between the movable window panel and the rail, such as to absorb or dampen or attenuate vibrations from the movable window panel **20** as the vehicle travels along the road. Further, the biasing element may provide a biasing force between at least one side of the movable window panel **20** along the upper edge region and/or the lower edge region and the rail of the frame **14**, such as to maintain alignment of the movable window panel **20** along the channels of the frame **14** as the movable window panel **20** moves between the closed position and the opened position and/or to maintain the movable window panel **20** in place after being moved to a position at or between the closed position and the opened position. That is, the rear slider window assembly includes a spring loaded track along which the movable window panel moves relative to the fixed window.

(23) Referring to FIGS. **2-9**, an upper rail **24** of the frame **14** defines an elongated, generally U-shaped channel portion that is disposed generally horizontally along the rear slider window and spanning at least part of each of the fixed window panels **16, 18**. The upper rail **24** may extend along an upper edge region of the fixed window panels **16, 18** and/or an upper applique **17** that extends between the fixed window panels and above the aperture. An upper edge region or portion **22** of the movable window panel **20** is movably received along the upper rail **24** of the frame **14**. For example, a guide element **26** is attached along the upper edge region **22** of the movable window panel **20** and is movably received along the U-shaped channel portion so as to be movable or slidable along the channel as the movable window panel **20** is moved between the closed position and the opened position relative to the upper rail **24** of the frame **14** and the one or more fixed window panels **16, 18**.

(24) The guide element **26** includes a receiving portion or trim portion **28** that receives and extends along the upper edge region **22** of the movable window panel **20**. The trim portion **28** may include an elongated, generally U-shaped rail defining a generally U-shaped channel for receiving the upper edge region **22** of the movable window panel **20**. The trim portion **28** of the guide element **26** may be attached to the movable window panel **20** in any suitable manner, such as via adhesive or press-fit at the upper edge region **22**. For example, the trim portion **28** may comprise a flexible metallic element that is disposed along the upper edge region **22** of the movable window panel **20** and at least partially compressed or crimped to secure the guide element to the movable window panel **20**.

(25) With the generally U-shaped guide element **26** attached along the upper edge region **22** of the movable window panel **20**, a first side **28a** of the trim portion **28** extends at least partially at and along a first side **20a** of the movable window panel **20** (e.g., an in-cabin facing side of the movable window panel) and a second side **28b** of the trim portion **28** extends at least partially at and along a second side **20b** of the movable window panel **20** opposite the first side **20a** (e.g., an exterior facing side of the movable window panel). At least one biasing element or spring element or absorbing element or damping element or attenuating element **30** is disposed at and along the first side **28a** of the trim portion **28** between the guide element and a first side or inner surface **24a** of the upper rail **24** and at least one biasing element **30** is disposed at and along the second side **28b** of the trim portion **28** between the guide element and a second side or inner surface **24b** of the upper rail **24** (e.g., FIGS. **2A** and **3A**). In the illustrated example, two biasing elements **30** are disposed at and along each of the first side **28a** and the second side **28b** of the guide element **26** and it should

be understood that the guide element **26** may include any suitable number of biasing elements **30** at and along the guide element **26**.

(26) The biasing elements **30** may comprise any suitable flexible and/or compressible material for absorbing or damping or attenuating vibrations between the upper rail **24** and movable window panel **20** and for applying the biasing force between the guide element **26** and the upper rail **24**. For example, the biasing elements **30** may include curved or arcuate metallic bands or leaf springs having respective ends fixed along the outer surface of the trim portion **28** of the guide element **26** and curving outward from the guide element **26** so that, when at least partially compressed between the upper rail **24** and the trim portion **28**, the metallic bands flatten and flex toward the trim portion **28**. The biasing elements **30** may be integrally formed with the guide element **26**, such that the curved metallic bands **30** extend from and are integrally formed with the metallic trim portion **28**. Optionally, the biasing elements **30** may include strips of compressible or flexible material, such as a rubber or plastic material, that are adhesively attached at and along the respective sides of the trim portion **28**. The strips of compressible material may include curved end portions, such as to allow slidable movement of the guide element **26** along the upper rail **24**.

(27) With the guide element **26** and upper edge region **22** of the movable window panel **20** received along the upper rail **24**, the biasing elements **30** are at least partially compressed between the respective inner surfaces of the upper rail **24** and the movable window panel **20**. Thus, the biasing elements **30** apply respective biasing forces along both sides of the movable window panel **20** and between the movable window panel **20** and the upper rail **24**, such as to urge the movable window panel **20** toward a centerline of the upper rail **24** (that extends parallel to the longitudinal axis of the upper rail **24** and is approximately equidistant to the first inner surface **24a** and the second inner surface **24b** of the upper rail **24**) as the movable window panel **20** moves or slides along the upper rail **24**. In some examples, the upper rail **24** may curve according to a curvature of the fixed window panels and the channel of the upper rail **24** may similarly curve so that the movable window panel **20** moves along a curved path of travel between the closed position and the opened position. The biasing elements **30** urge the movable window panel **20** toward the center of the channel so that, as the window panel **20** is moved, the window panel **20** follows the curved path of travel and maintains a generally consistent or constant distance from the upper rail and/or fixed window panels.

(28) Optionally, the biasing elements **30** apply a frictional force along the upper rail **24**, such that, when the drive system and/or the user moves the window panel **20** between the closed position and the opened position, the frictional force is overcome to move the window panel **20** and, when the drive system and/or user stops moving the window panel **20**, the frictional force between the biasing elements **30** and the upper rail **24** maintains the window panel **20** in position along the rail. In other words, the biasing elements **30** are compressed between the upper rail **24** and the movable window panel **20** to at least partially resist movement of the movable window panel **20** along the upper rail **24** when a moving force (e.g., from the drive system or a manual force from the user) is not being applied to move the window panel **20**.

(29) FIGS. 2A-2C and 3A depict sectional views of the upper edge region **22** of the movable window panel **20** and the guide element **26** received along the U-shaped channel of the upper rail **24**. In the illustrated example of FIGS. 2A and 3A, the biasing elements **30** are disposed at and along respective end regions of the guide element **26** and thus, at a central portion of the guide element **26**, the trim portion **28** engages the upper edge portion **22** of the movable window panel **20** so that the first side **28a** extends at least partially along the first side **20a** of the movable window panel **20** and the second side **28b** extends at least partially along the second side **20b** of the movable window panel **20** and no biasing element **30** extends between the trim portion **28** and the upper rail **24**. As shown in FIGS. 2B and 3A, the biasing elements **30** are curved so that a central portion of the respective biasing elements **30** extends from the trim portion **28** and engages the inner surface of the upper rail **24**. Respective end portions or outboard portions of the biasing

elements **30** may not engage the inner surface of the upper rail **24** when the movable window panel **20** is centered along the rail **24** (e.g., FIGS. 2C and 3A). As the biasing elements **30** are compressed toward one side of the rail **24** (e.g., a force is applied at one side of the movable window panel), the curved biasing element **30** may flatten and an engagement area between the biasing element **30** and the inner surface of the rail **24** increases.

(30) Optionally, the biasing element may be attached directly to the surface of the movable window panel. For example, and referring to FIGS. 10-17, a rear slider window assembly **110** includes a movable window panel **120** having an upper edge region or portion **122** that is movably received along the channel of an upper rail **124** so as to be movable or slidable along the upper rail **124** relative to fixed window panel portions **116**, **118**. One or more absorbing elements or damping elements or attenuating elements or biasing elements **130** (e.g., two or more biasing elements) are attached at the upper edge region **122** at and along a first side **120a** of the movable window panel **120** (e.g., an in-cabin facing side of the movable window panel) and one or more biasing elements **130** (e.g., two or more biasing elements) are attached at the upper edge region **122** at and along a second side **120b** of the movable window panel **120** opposite the first side **120a** (e.g., an exterior facing side of the movable window panel). For example, the biasing elements **130** may be adhesively attached at the respective surfaces of the movable window panel **120** or the biasing elements **130** may be attached via threaded fasteners. Optionally, the movable window panel **120** and/or the biasing elements may be locally deformed (e.g., heated so as to at least partially melt or deform) to join the biasing elements **130** at the movable window panel **120**.

(31) Thus, with the upper edge region **122** and the biasing elements **130** received along the upper rail **124**, the biasing elements **130** are disposed, respectively, between the first side **120a** of the movable window panel **120** and a first side or inner surface **124a** of the upper rail **124** and the second side **120b** of the movable window panel **120** and a second side or inner surface **124b** of the upper rail **124**. The biasing elements **130** may be at least partially compressed between the respective inner surfaces of the upper rail **124** and the movable window panel **120**, such as to align the window panel **120** with the centerline of the rail **124** or maintain a position of the window panel **120** along the rail **124**.

(32) In the illustrated example, the biasing elements **130** include arcuate spring elements that attach at opposing ends to the movable window panel **120** and that curve away from the movable window panel **120** between the opposing ends to allow for flexing of the biasing element **130** toward the movable window panel **120**. For example, and as shown in FIGS. 14-17, each biasing element **130** includes a curved central portion **132** and flat or planar tabs **134** extending from opposite ends of the curved portion **132**. The tabs **134** are attached at the outer surface of the movable window panel **120**, such as adhesively attached, and the curved portion **132** extends at least partially away from the outer surface of the window panel **120** to engage the inner surface of the upper rail **124**. The tabs **134** may include a plurality of projections that engage the window panel and establish a bond line thickness for the adhesive used to adhere or bond the tabs and biasing element to the window panel.

(33) FIGS. 10A-10C and 11A depict sectional views of the upper edge region **122** of the movable window panel **120** and the biasing elements **130** disposed at and along the upper edge region **122**. As shown in FIGS. 10A and 11A, the tabs **134** of the biasing elements **130** respectively are attached at and extend along the first side **120a** and the second side **120b** of the movable window panel **120**. As shown in FIGS. 10B and 11A, the curved central portion **132** extends away from the movable window panel **120** and engages the inner surface of the upper rail **124**. The curved portion **132** allows the biasing element **130** to flex or flatten when compressed between the window panel **120** and the rail **124**, such that a portion of the curved portion **132** may be disposed between the window panel **120** and the upper rail **124** and not engaging the upper rail **124** (e.g., FIGS. 10C and 11A). As the biasing elements **130** are compressed toward one side of the rail **124** (e.g., a force is applied at one side of the movable window panel), the curved portion **132** of the biasing element **30**

may flatten and an engagement area between the curved portion **132** and the inner surface of the rail **124** increases.

(34) Thus, the rear slider window assembly at the rear portion of the cabin of the vehicle includes a fixed window panel and a movable window panel that is movable or slidable relative to the fixed window panel along a frame of the window assembly. The frame includes an upper channel or rail and an upper edge region of the movable window panel is movably received along the upper channel. A first absorbing element or damping element or attenuating element or biasing element is disposed between a first side of the movable window panel (e.g., an in-cabin facing side of the movable window panel) and an inner surface of the channel and a second absorbing element or damping element or attenuating element or biasing element is disposed between a second side of the movable window panel (e.g., an exterior facing side of the movable window panel) and an inner surface of the channel. For example, the biasing elements are disposed along a guide element or trim piece that is attached along the upper edge region of the movable window panel or the biasing elements are attached to the respective surfaces of the movable window panel (e.g., adhesively attached). The biasing elements absorb or dampen or attenuate vibrations during travel of the vehicle and bias or urge the movable window panel toward a center of the channel to maintain alignment of the movable window panel along the upper rail as the movable window panel moves between an opened position and a closed position.

(35) The window panel may comprise a glass window panel, such as transparent tempered glass window panel or the like. Optionally, the window panel may comprise a non-glass material, such as a transparent or substantially transparent plastic or polymeric or acrylic material or the like. Although shown and described as a vehicular rear slider window assembly configured to be disposed at the rear portion of a cabin of a pickup truck, the window assembly may be suitable for other rear windows (such as a liftgate window such as for a hatchback vehicle or van or the like). Further, although described as having the biasing elements disposed between the movable window panel and the frame at the upper rail, the biasing elements may be disposed between the movable window panel and the frame at the lower rail (or optionally, the lower region of the movable window panel may be disposed at a carriage that is slidably received within the lower rail). Moreover, the biasing elements may be attached at the rail with the movable window panel moving along the rail relative to the fixed window panels and the biasing elements.

(36) Optionally, the window assembly or assemblies and drive system may utilize aspects of the window assemblies and drive systems described in U.S. Pat. Nos. 11,331,984; 10,501,977; 8,938,914; 8,915,018; 8,881,458; 8,402,695; 7,073,293; 7,003,916; 6,691,464 and/or 5,799,444, and/or U.S. Publication Nos. US-2021/0355737; US-2014/0047772; US-2008/0127563; US-2006/0107600; US-2004/0020131 and/or US-2003/0213179, all of which are hereby incorporated herein by reference in their entireties.

(37) Changes and modifications to the specifically described embodiments may be carried out without departing from the principles of the present invention, which is intended to be limited only by the scope of the appended claims, as interpreted according to the principles of patent law.

Claims

1. A vehicular rear slider window assembly, the vehicular rear slider window assembly comprising: a fixed window panel, wherein the fixed window panel comprises an opening; a frame portion fixed relative to the fixed window panel, wherein the frame portion comprises an upper rail; a movable window panel, wherein the movable window panel comprises a first side and a second side, and wherein the second side of the movable window panel is opposite the first side; wherein an upper edge region of the movable window panel is movably received in a channel of the upper rail, and wherein the channel of the upper rail comprises a first inner surface and a second inner surface, and wherein the first inner surface and the second inner surface are along opposing sides of

the channel; wherein the movable window panel is movable along the upper rail between (i) a closed position where the movable window panel is disposed at the opening and (ii) an opened position where the movable window panel is disposed at least partially along the fixed window panel; a first biasing element disposed between the first side of the movable window panel along the upper edge region and the first inner surface of the upper rail along the channel; wherein the first biasing element comprises (i) a pair of first attaching portions that are disposed at the first side of the upper edge region of the movable window panel and that are spaced from one another at the first side of the movable window panel and (ii) a first biasing portion that extends between the pair of first attaching portions and that is spaced from the first side of the movable window panel; a second biasing element disposed between the second side of the movable window panel along the upper edge region and the second inner surface of the upper rail along the channel; wherein the second biasing element comprises (i) a pair of second attaching portions that are disposed at the upper edge region of the second side of the movable window panel and that are spaced from one another at the second side of the movable window panel and (ii) a second biasing portion that extends between the pair of second attaching portions and that is spaced from the second side of the movable window panel; wherein the first biasing portion of the first biasing element is at least partially compressed between the first inner surface of the upper rail and the first side of the movable window panel and the first biasing element urges the movable window panel away from the first inner surface of the upper rail; wherein the second biasing portion of the second biasing element is at least partially compressed between the second inner surface of the upper rail and the second side of the movable window panel and the second biasing element urges the movable window panel away from the second inner surface of the upper rail; and wherein, as the movable window panel is moved between the closed position and the opened position, the first biasing element and the second biasing element guide movement of the movable window panel in a direction parallel to a centerline of the channel of the upper rail.

2. The vehicular rear slider window assembly of claim 1, wherein the pair of first attaching portions of the first biasing element are adhesively attached at the first side of the movable window panel, and wherein the pair of second attaching portions of the second biasing element are adhesively attached at the second side of the movable window panel.

3. The vehicular rear slider window assembly of claim 1, wherein a guide element is attached to the movable window panel at the upper edge region, and wherein the first biasing element is disposed at a first side of the guide element at the first side of the movable window panel and the second biasing element is disposed at a second side of the guide element at the second side of the movable window panel.

4. The vehicular rear slider window assembly of claim 3, wherein the guide element is adhesively attached to the movable window panel at the upper edge region.

5. The vehicular rear slider window assembly of claim 3, wherein the first biasing element and the second biasing element are integrally formed with the guide element.

6. The vehicular rear slider window assembly of claim 3, wherein the first biasing element and the second biasing element are adhesively attached to the guide element.

7. The vehicular rear slider window assembly of claim 1, wherein the first biasing element comprises a first arcuate spring element that extends from the first side of the movable window panel and engages the first inner surface of the upper rail, and wherein the second biasing element comprises a second arcuate spring element that extends from the second side of the movable window panel and engages the second inner surface of the upper rail.

8. The vehicular rear slider window assembly of claim 1, wherein the first biasing element and the second biasing element comprise one selected from the group consisting of (i) a flexible metallic material, (ii) a rubber material and (iii) a plastic material.

9. The vehicular rear slider window assembly of claim 1, wherein the channel comprises a generally U-shaped channel, and wherein the first inner surface of the upper rail faces the second

inner surface of the upper rail.

10. The vehicular rear slider window assembly of claim 1, wherein the first biasing element and the second biasing element at least partially attenuate vibrations from passing between the upper rail and the movable window panel.

11. The vehicular rear slider window assembly of claim 1, wherein the first biasing element and the second biasing element cooperate to urge the movable window panel toward the centerline of the channel of the upper rail as the movable window panel is moved between the closed position and the opened position.

12. The vehicular rear slider window assembly of claim 11, wherein the centerline is equidistant to the first inner surface of the upper rail and the second inner surface of the upper rail.

13. The vehicular rear slider window assembly of claim 11, wherein the centerline is parallel to a longitudinal axis of the upper rail.

14. The vehicular rear slider window assembly of claim 1, wherein the vehicular rear slider window assembly is configured to be disposed at a rear portion of a cabin of a pickup truck.

15. A vehicular rear slider window assembly, the vehicular rear slider window assembly comprising: a fixed window panel, wherein the fixed window panel comprises an opening; a frame portion fixed relative to the fixed window panel, wherein the frame portion comprises an upper rail; a movable window panel, wherein the movable window panel comprises a first side and a second side, and wherein the second side of the movable window panel is opposite the first side; wherein an upper edge region of the movable window panel is movably received in a channel of the upper rail, and wherein the channel of the upper rail comprises a first inner surface and a second inner surface, and wherein the first inner surface and the second inner surface are along opposing sides of the channel; wherein the movable window panel is movable along the upper rail between (i) a closed position where the movable window panel is disposed at the opening and (ii) an opened position where the movable window panel is disposed at least partially along the fixed window panel; a first biasing element disposed at the upper edge region and between the first side of the movable window panel and the first inner surface of the upper rail along the channel; wherein the first biasing element is adhesively attached at the first side of the movable window panel, and wherein the first biasing element comprises (i) a pair of first attaching portions that are adhesively attached at the upper edge region of the first side of the movable window panel and that are spaced from one another at the first side of the movable window panel and (ii) a first arcuate spring element that extends between the pair of first attaching portions and that is spaced from the first side of the movable window panel, and wherein the first arcuate spring element engages the first inner surface of the upper rail; a second biasing element disposed at the upper edge region and between the second side of the movable window panel and the second inner surface of the upper rail along the channel; wherein the second biasing element is adhesively attached at the second side of the movable window panel, and wherein the second biasing element comprises (i) a pair of second attaching portions that are adhesively attached at the upper edge region of the second side of the movable window panel and that are spaced from one another at the second side of the movable window panel and (ii) a second arcuate spring element that extends between the pair of second attaching portions and that is spaced from the second side of the movable window panel, and wherein the second arcuate spring element engages the second inner surface of the upper rail; wherein the first arcuate spring element of the first biasing element is at least partially compressed between the first inner surface of the upper rail and the first side of the movable window panel and the first biasing element urges the movable window panel away from the first inner surface of the upper rail; wherein the second arcuate spring element of the second biasing element is at least partially compressed between the second inner surface of the upper rail and the second side of the movable window panel and the second biasing element urges the movable window panel away from the second inner surface of the upper rail; and wherein, as the movable window panel is moved between the closed position and the opened position, the first biasing element and the

second biasing element guide movement of the movable window panel in a direction parallel to a centerline of the channel of the upper rail.

16. The vehicular rear slider window assembly of claim 15, wherein the first biasing element and the second biasing element comprise one selected from the group consisting of (i) a flexible metallic material, (ii) a rubber material and (iii) a plastic material.

17. The vehicular rear slider window assembly of claim 15, wherein the first biasing element and the second biasing element cooperate to urge the movable window panel toward the centerline of the channel of the upper rail as the movable window panel is moved between the closed position and the opened position.

18. The vehicular rear slider window assembly of claim 15, wherein the vehicular rear slider window assembly is configured to be disposed at a rear portion of a cabin of a pickup truck.

19. A vehicular rear slider window assembly, the vehicular rear slider window assembly comprising: a fixed window panel, wherein the fixed window panel comprises an opening; a frame portion fixed relative to the fixed window panel, wherein the frame portion comprises an upper rail; a movable window panel, wherein the movable window panel comprises a first side and a second side, and wherein the second side of the movable window panel is opposite the first side; wherein an upper edge region of the movable window panel is movably received in a channel of the upper rail, and wherein the channel of the upper rail comprises a first inner surface and a second inner surface, and wherein the first inner surface and the second inner surface are along opposing sides of the channel; wherein the movable window panel is movable along the upper rail between (i) a closed position where the movable window panel is disposed at the opening and (ii) an opened position where the movable window panel is disposed at least partially along the fixed window panel; a guide element attached to the movable window panel at the upper edge region, wherein the guide element comprises (i) a first side at the first side of the movable window panel and (ii) a second side at the second side of the movable window panel; a first biasing element disposed at the first side of the guide element and between the first side of the movable window panel and the first inner surface of the upper rail along the channel; wherein the first biasing element comprises (i) a pair of first attaching portions that are disposed at the first side of the guide element at the upper edge region of the movable window panel and that are spaced from one another at the first side of the guide element and (ii) a first arcuate spring element that extends between the pair of first attaching portions and that is spaced from the first side of the guide element, and wherein the first arcuate spring element engages the first inner surface of the upper rail; a second biasing element disposed at the second side of the guide element and between the second side of the movable window panel and the second inner surface of the upper rail along the channel; wherein the second biasing element comprises (i) a pair of second attaching portions that are disposed at the second side of the guide element at the upper edge region of the movable window panel and that are spaced from one another at the second side of the guide element and (ii) a second arcuate spring element that extends between the pair of second attaching portions and that is spaced from the second side of the guide element, and wherein the second arcuate spring element engages the second inner surface of the upper rail; wherein the first arcuate spring element of the first biasing element is at least partially compressed between the first inner surface of the upper rail and the first side of the guide element and the first biasing element urges the movable window panel away from the first inner surface of the upper rail; wherein the second arcuate spring element of the second biasing element is at least partially compressed between the second inner surface of the upper rail and the second side of the guide element and the second biasing element urges the movable window panel away from the second inner surface of the upper rail; and wherein, as the movable window panel is moved between the closed position and the opened position, the first biasing element and the second biasing element guide movement of the movable window panel in a direction parallel to a centerline of the channel of the upper rail.

20. The vehicular rear slider window assembly of claim 19, wherein the guide element is

adhesively attached to the movable window panel at the upper edge region.

21. The vehicular rear slider window assembly of claim 19, wherein the first biasing element and the second biasing element are integrally formed with the guide element.

22. The vehicular rear slider window assembly of claim 19, wherein the pair of first attaching portions of the first biasing element are adhesively attached at the first side of the guide element, and wherein the pair of second attaching portions of the second biasing element are adhesively attached at the second side of the guide element.

23. The vehicular rear slider window assembly of claim 19, wherein the first biasing element and the second biasing element comprise one selected from the group consisting of (i) a flexible metallic material, (ii) a rubber material and (iii) a plastic material.

24. The vehicular rear slider window assembly of claim 19, wherein the first biasing element and the second biasing element cooperate to urge the movable window panel toward the centerline of the channel of the upper rail as the movable window panel is moved between the closed position and the opened position.

25. The vehicular rear slider window assembly of claim 19, wherein the vehicular rear slider window assembly is configured to be disposed at a rear portion of a cabin of a pickup truck.
